

Java

1. Basic reflection

Procedure 1.1 (Enumerating methods of class)

```
Class clazz = String.class; // by class name
Class clazz = instance.getClass(); // get
from an instance

for(java.lang.reflect.Method m :
clazz.getMethods()) {
    System.out.println(m);
}
```

2. Common packages

- `java.util` — collections (List, Map, Set, ArrayList, HashMap, Scanner, Random, etc.)
- `java.io` — file/stream I/O (File, BufferedReader, IOException)
- `java.lang` — core classes, **auto-imported already** (String, Object, Math, Integer, etc. — no import needed)
- `java.lang.reflect` — reflection (Method, Field, Modifier)
- `java.time` — dates/times (LocalDate, LocalDateTime, Duration)
- `java.util.stream` — streams (Stream, Collectors)
- `java.util.concurrent` — threading utilities

3. Footguns

- `==` tests pointer equality, not value equality, if the arguments are objects
 - Use `equals` for value equality on common classes
- Similarly, don't use `<`, `>`, `<=`, `>=` on objects
 - Use `compareTo`, which returns positive value if LHS > RHS, negative value if LHS < RHS, and 0 if equal